BTS-I(R)-11-23-2191 Reg. No. 2 0 6 2 3 0 4 2

C

B.Tech. Degree I Semester Regular Examination November 2023

CE/ME/SE 23-200-0102A ENGINEERING CHEMISTRY

(2023 *Scheme*)

Time: 3 Hours Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Explain the basic concepts of chemical thermodynamics and quantum chemistry.
CO2: Illustrate the spectroscopic methods in characterizing materials.
CO3: Develop electro chemical methods to protect different metals from corrosion.
CO4: Interpret the chemistry of a few important engineering materials and their industrial applications.
CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

			$(5 \times 2 = 10)$	Marks	BL	CO	PO
I.	(a)	Explain Chemical potential and fugacity.		2	L1	1	3
	(b)	The wavefunction $\Psi = A \cos x$ for $0 < x < \pi/2$. Find the value of A.		2	L3	5	2
	(c)	State and explain Beer-Lambert's law.		2	L2	2	3
	(d)	Write a note on the properties of graphene.		2	L1	4	2
	(e)	Explain the working of Lead storage battery.		2	L1	3	2

PART B

			$(4 \times 10 = 40)$				
II.		Derive van't Hoff reaction isotherm.		10	L2	1	3
		OR					
III.		Explain simple eutectic system with a neat labelled phase diagram.		10	L2	1	3
IV.		Derive operator form of Schrodinger wave equation from classical wave equation.		10	L2	1	3
		OR					
V.		An electron is confined to move in one dimensional box of length 'a'. Derive its ground state using quantum mechanical principles.		10	L2	1	3
VI.	(a)	Draw the low and high resolution ^1H NMR spectra of (i) CH_3-CH_2-CH-Cl_2 (ii) Ethanol (iii) CH_3-O- CH_3		6	L3	2	2
	(b)	Prove that the distance between consecutive lines in a rotational spectrum is 2B.		4	L2	2	3
		OR					
VII.	(a)	Explain the basic principle of NMR spectra. Indicate using a sketch the chemical shift and spin-spin splitting in the NMR spectrums of the molecule CH_3-CHO.		6	L3	2	3
	(b)	Write the principle behind IR spectroscopy.		4	L2	2	3
VIII.	(a)	What are Refractories? Discuss their properties and classification.		5	L1	4	3
	(b)	Derive Nernst equation.		5	L1	3	3
		OR					
IX.	(a)	What are lubricants? Discuss their classification and applications.		5	L1	4	3
	(b)	Give a brief account of methods of corrosion prevention.		5	L1	3	3

Bloom's Taxonomy Levels

L1 = 38.88%, L2 = 44.44%, L3 = 16.7%.

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